MIL-W-23351/4B
20 October 1977
SUPERSEDING
MIL-W-23351/4A
10 October 1966

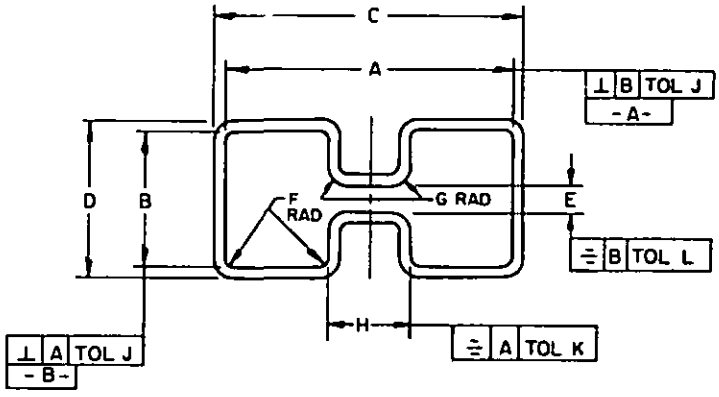
MILITARY SPECIFICATION SHEET

WAVEGUIDES, DOUBLE RIDGE

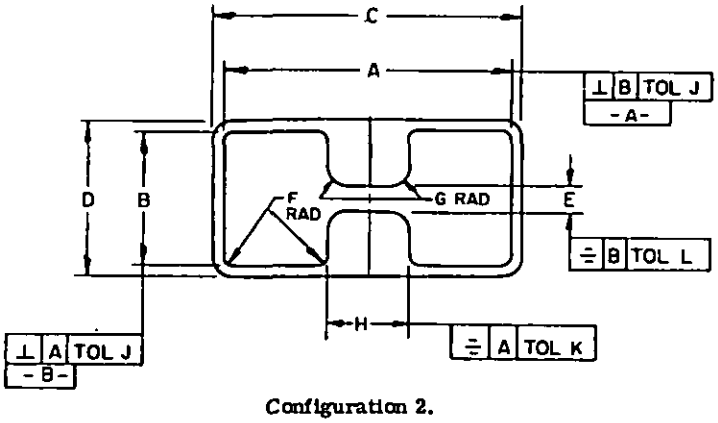
(BANDWIDTH RATIO 2.4:1)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the waveguides described herein shall consist of this document and the latest issue of Specification MIL-W-23351.



Configuration 1.



Configuration 2.

FIGURE 1. Waveguide.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, Department of the Navy, ATTN: ELEX 5043, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FSC 5985

TABLE I. Part number dimensions and material.

Part number M23351/4-	Dimensions 1/ A	B	C	D	E
001 2/ 002 003 004	29.667 ±.020 (753.54) (.51)	13.795 ±.020 (350.39) (.51)	29.917 Max (759.89)	14.045 Max (356.74)	5.863 ±.010 (148.92) (.25)
005 006 007 008	19.428 ±.015 (493.47) (.38)	9.034 ±.015 (229.46) (.38)	19.678 Max (499.82)	9.284 Max (235.81)	3.839 ±.007 (97.51) (.18)
009 010 011 012	12.333 ±.010 (313.26) (.25)	5.737 ±.010 (145.72) (.25)	12.583 ±.015 (319.61) (.38)	5.987 ±.015 (152.07) (.38)	2.437 ±.005 (61.90) (.13)
013 014 015 016 2/	8.100 ±.010 (205.74) (.25)	3.767 ±.010 (95.68) (.25)	8.350 ±.015 (212.09) (.38)	4.017 ±.015 (102.03) (.38)	1.601 ±.004 (40.67) (.10)
017 3/ 018 019 020	6.167 ±.005 (156.64) (.13)	2.868 ±.005 (72.85) (.13)	6.417 ±.005 (162.99) (.13)	3.118 ±.005 (79.20) (.13)	1.219 ±.003 (30.96) (.08)
021 022 023 024	3.455 ±.004 (87.76) (.10)	1.607 ±.004 (40.82) (.10)	3.615 ±.004 (91.82) (.10)	1.767 ±.004 (44.88) (.10)	.6830 ±.003 (17.348) (.08)
025 026 027 028	2.590 ±.004 (65.79) (.10)	1.205 ±.004 (30.61) (.10)	2.750 ±.004 (69.85) (.10)	1.365 ±.004 (34.67) (.10)	.512 ±.002 (13.00) (.05)
029 030 031 032	1.480 ±.003 (37.59) (.08)	.688 ±.003 (17.48) (.08)	1.608 ±.004 (40.84) (.10)	.816 ±.004 (20.73) (.10)	.2920 ±.002 (7.417) (.05)
033 034 035 036	1.090 ±.003 (27.69) (.08)	.506 ±.003 (12.85) (.08)	1.190 ±.003 (30.23) (.08)	.606 ±.003 (15.39) (.08)	.215 ±.002 (5.46) (.05)
037 038 039 040	.691 ±.003 (17.55) (.08)	.321 ±.003 (8.15) (.08)	.791 ±.003 (20.09) (.08)	.421 ±.003 (10.69) (.08)	.136 ±.002 (3.45) (.05)
041 042 043 044	.471 ±.003 (11.96) (.08)	.219 ±.003 (5.56) (.08)	.551 ±.003 (14.00) (.08)	.299 ±.003 (7.59) (.08)	.0930 ±.002 (2.362) (.05)
045 046 047 048 3/	.288 ±.003 (7.32) (.08)	.134 ±.003 (3.40) (.08)	.368 ±.003 (9.35) (.08)	.214 ±.003 (5.44) (.08)	.0570 ±.002 (1.448) (.05)
049 3/ 4/ 050 3/ 4/ 051 3/ 4/ 052 3/ 4/	1.025 ±.003 (26.04) (.08)	.475 ±.003 (12.06) (.08)	1.125 ±.005 (28.58) (.13)	.575 ±.005 (14.60) (.13)	.191 ±.003 (4.85) (.08)

See footnotes at end of table.

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TABLE I. Part number dimensions and material - Continued.

Part number M23351/4-	Dimensions (continued) 1/ F Max	G ±10%	H	J	K	L	Material
001 2/ 002 003 004	.063 (1.60)	1.173 (29.79)	7.417 ±.010 (188.39) (.25)	.020 (.51)	.010 (.25)	.010 (.25)	Aluminum alloy Copper alloy Copper Silver alloy
005 006 007 008	.050 (1.27)	.768 (19.50)	4.857 ±.007 (123.37) (.18)	.015 (.38)	.007 (.18)	.007 (.18)	Aluminum alloy Copper alloy Copper Silver alloy
009 010 011 012	.050 (1.27)	.487 (12.37)	3.083 ±.005 (78.31) (.13)	.010 (.25)	.005 (.13)	.005 (.13)	Aluminum alloy Copper alloy Copper Silver alloy
013 014 015 016 2/	.050 (1.27)	.320 (8.13)	2.025 ±.004 (51.44) (.10)	.010 (.25)	.004 (.10)	.004 (.10)	Aluminum alloy Copper alloy Copper Silver alloy
017 3/ 018 019 020	.050 (1.27)	.244 (6.20)	1.542 ±.003 (39.17) (.08)	.005 (.13)	.003 (.08)	.003 (.08)	Aluminum alloy Copper alloy Copper Silver alloy
021 022 023 024	.050 (1.27)	.137 (3.48)	.8640 ±.003 (21.946) (.08)	.004 (.10)	.003 (.08)	.003 (.08)	Aluminum alloy Copper alloy Copper Silver alloy
025 026 027 028	.050 (1.27)	.102 (2.59)	.648 ±.002 (16.46) (.05)	.004 (.10)	.002 (.05)	.002 (.05)	Aluminum alloy Copper alloy Copper Silver alloy
029 030 031 032	.030 (.76)	.058 (1.47)	.3700 ±.002 (9.398) (.05)	.003 (.08)	.002 (.05)	.002 (.05)	Aluminum alloy Copper alloy Copper Silver alloy
033 034 035 036	.030 (.76)	.043 (1.09)	.272 ±.002 (6.91) (.05)	.003 (.08)	.002 (.05)	.002 (.05)	Aluminum alloy Copper alloy Copper Silver alloy
037 038 039 040	.020 (.51)	.027 (.69)	.173 ±.002 (4.39) (.05)	.003 (.08)	.002 (.05)	.002 (.05)	Aluminum alloy Copper alloy Copper Silver alloy
041 042 043 044	.015 (.38)	.019 (.48)	.1180 ±.002 (2.997) (.05)	.003 (.08)	.002 (.05)	.002 (.05)	Aluminum alloy Copper alloy Copper Silver alloy
045 046 047 048 3/	.015 (.38)	.011 (.28)	.0720 ±.002 (1.829) (.05)	.003 (.08)	.002 (.05)	.002 (.05)	Aluminum alloy Copper alloy Copper Silver alloy
049 3/ 4/ 050 3/ 4/ 051 3/ 4/ 052 3/ 4/	.030 (.76)	.062 (1.57)	.256 ±.003 (6.50) (.08)	.003 (.08)	.003 (.08)	.003 (.08)	Aluminum alloy Copper alloy Copper Silver alloy

1/ Dimensions are in inches. Metric equivalents (in parentheses) are given for general information only and are based upon 1.00 inch = 25.4 mm.
2/ The waveguide shall be in accordance with configuration 1.
3/ The waveguide shall be in accordance with configuration 2.
4/ Eccentricity shall be .004 maximum.

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REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1 and table I.
Material: See table I.
Part number: M23351/4- (dash number from table I).

NOTES:

Engineering information and frequency range: See table II.
Cross reference: See table III.

TABLE II. Engineering information.

Part number M23351/4-	Suggested frequency range TE_{10} mode	Cutoff frequency		Theoretical attenuation at $f = \sqrt{3}\, f_{c_{10}}$ 1/ 2/	Theoretical power handling capacity at $f = \sqrt{3}\, f_{c_{10}}$ 3/
		TE_{10} mode	TE_{20} mode		
	GHz	GHz	GHz	dB/ft	kW
001-XXX 002-XXX 003-XXX 004-XXX	0.175- 0.420	0.146	0.430	0.00035 0.00034 0.00023 0.00025	61,960
005-XXX 006-XXX 007-XXX 008-XXX	0.267- 0.640	0.222	0.657	0.00065 0.00064 0.00043 0.00046	26,570
009-XXX 010-XXX 011-XXX 012-XXX	0.420- 1.000	0.350	1.034	0.00128 0.00126 0.00085 0.00091	10,710
013-XXX 014-XXX 015-XXX 016-XXX	0.640- 1.530	0.533	1.575	0.0024 0.0024 0.0016 0.0017	4,620
017-XXX 018-XXX 019-XXX 020-XXX	0.840- 2.000	0.700	2.068	0.0036 0.0036 0.0024 0.0026	2,676
021-XXX 022-XXX 023-XXX 024-XXX	1.500- 3.600	1.249	3.692	0.0087 0.0086 0.0058 0.0062	840.5
025-XXX 026-XXX 027-XXX 028-XXX	2.000- 4.800	1.666	4.925	0.0134 0.0132 0.0089 0.0095	472.5
029-XXX 030-XXX 031-XXX 032-XXX	3.500- 8.200	2.915	8.620	0.0307 0.0303 0.0204 0.0218	151.3
033-XXX 034-XXX 035-XXX 036-XXX	4.750-11.000	3.961	11.705	0.0487 0.0481 0.0324 0.0347	83.72
037-XXX 038-XXX 039-XXX 040-XXX	7.500-18.000	6.239	18.464	0.0964 0.0951 0.0641 0.0686	33.58

See footnotes at end of table.

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TABLE II. Engineering information - Continued.

Part number M23351/4-	Suggested frequency range TE_{10} mode	Cutoff frequency		Theoretical attenuation at $f = \sqrt{3}\, f_{c_{10}}$ 1/ 2/	Theoretical power handling capacity at $f = \sqrt{3}\, f_{c_{10}}$ 3/
		TE_{10} mode	TE_{20} mode		
	GHz	GHz	GHz	dB/ft	kW
041-XXX	11.000-26.500	9.363	27.080	0.171	15.63
042-XXX				0.169	
043-XXX				0.114	
044-XXX				0.122	
045-XXX	18.000-40.000	14.995	44.285	0.358	5.834
046-XXX				0.353	
047-XXX				0.238	
048-XXX				0.255	
049-XXX	4.75 -11.00	4.086	12.07	---	---
050-XXX					
051-XXX					
052-XXX					

1/ $f_{c_{10}}$ = cutoff frequency for TE_{10} mode.

2/ Actual attenuation of waveguides may be considerably greater depending upon operating frequency and temperature. For further information or to calculate for frequencies other than those tabulated, see S. Hopfer, "The Design of Ridged Waveguides", IRE Transactions on Microwave Theory and Techniques, MTT-3 pp 20-29, October 1955, and MIL-HDBK-216, "Transmission Lines and Fittings", chapter 6. Typical resistivities of materials (at 20° C): aluminum alloy (6061) = 4.0 microhms-cm; copper alloy (brass) = 3.9 microhm-cm; copper - 1.77 microhm-cm; silver alloy (coin silver) = 2.03 microhm-cm.

3/ Actual power handling capability of waveguides may be considerably less depending upon waveguide temperature, altitude, operating frequency, etc. For further information or to calculate values for frequencies other than those tabulated, see S. Hopfer, "The Design of Ridged Waveguides", IRE Transactions on Microwave Theory and Techniques, MTT-3 pp 20-29, October 1955, and MIL-HDBK-216, "Transmission Lines and Fittings," chapter 6. The tabulated values were calculated based upon the E field produced breakdown in a nonpressurized air dielectric waveguide under continuous wave (CW) conditions. The breakdown strength of air was considered to be 15000 volts per centimeter, corner radii considered.

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TABLE III. Cross reference (type designation to part number).

Type designation	Part number M23351/4-	Type designation	Part number M23351/4-
WRD175U24A	001	WRD200D24A	025
WRD175U24B	002	WRD200D24B	026
WRD175U24C	003	WRD200D24C	027
WRD175U24S	004	WRD200D24S	028
WRD267U24A	005	WRD350D24A	029
WRD267U24B	006	WRD350D24B	030
WRD267U24C	007	WRD350D24C	031
WRD267U24S	008	WRD350D24S	032
WRD420U24A	009	WRD475D24A	033
WRD420U24B	010	WRD475D24B	034
WRD420U24C	011	WRD475D24C	035
WRD420U24S	012	WRD475D24S	036
WRD640U24A	013	WRD750D24A	037
WRD640U24B	014	WRD750D24B	038
WRD640U24C	015	WRD750D24C	039
WRD640U24S	016	WRD750D24S	040
WRD840U24A	017	WRD110C24A	041
WRD840U24B	018	WRD110C24B	042
WRD840U24C	019	WRD110C24C	043
WRD840U24S	020	WRD110C24S	044
WRD150D24A	021	WRD180C24A	045
WRD150D24B	022	WRD180C24B	046
WRD150D24C	023	WRD180C24C	047
WRD150D24S	024	WRD180C24S	048

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army -
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AR
Navy - MC, AS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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